

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

**Perihal Produk:** Sodium formaldehydesulfoxylate dihydrate  
**Product Description:** Sodium formaldehydesulfoxylate dihydrate  
**Cat No. :** 149190000; 149190010; 149190500  
**CAS No** 6035-47-8  
**Molecular Formula** C H3 Na O3 S . 2 H2 O

**Relevant identified uses of the substance or mixture and uses advised against**

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

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**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                        |                    |
|------------------------|--------------------|
| Germ Cell Mutagenicity | Category 2 (H341)  |
| Reproductive Toxicity  | Category 2 (H361d) |

**Label Elements**

**Signal Word**
**Warning**
**Hazard Statements**

H341 - Suspected of causing genetic defects  
 H361d - Suspected of damaging the unborn child

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## Precautionary Statements

### Prevention

- P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

- P308 + P313 - IF exposed or concerned: Get medical advice/attention

### Storage

- P403 - Store in a well-ventilated place

### Disposal

- P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

- EUH032 - Contact with acids liberates very toxic gas  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                                | CAS No    | Weight % |
|------------------------------------------|-----------|----------|
| Sodium formaldehydesulfoxylate dihydrate | 6035-47-8 | >95      |
| Sodium formaldehyde sulfoxylate          | 149-44-0  | -        |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

|                                    |                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | If symptoms persist, call a physician.                                                                                                           |
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| Ingestion                          | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                            |
| Inhalation                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| Self-Protection of the First Aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

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## Extinguishing media

### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

### **Extinguishing media which must not be used for safety reasons**

No information available.

## Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

## **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Sulfur oxides, Formaldehyde.

## Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

### Environmental precautions

Should not be released into the environment.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. Ensure adequate ventilation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep at temperatures below 40°C.

### Specific End Uses

Use in laboratories.

## **SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

### Control Parameters

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## Exposure Controls

### **Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### **Personal protective equipment**

#### **Eye Protection**

Wear safety glasses with side shields (or goggles)

#### **Hand Protection**

Protective gloves

#### **Skin and body protection**

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### **Recommended Filter type:**

Particulates filter conforming to EN 143

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice

### **Environmental exposure controls**

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### **Appearance**

#### **Physical State**

Solid

#### **Odor**

No information available

#### **Odor Threshold**

No data available

#### **pH**

No information available

10% aq. solution

#### **Melting Point/Range**

63 - 64 °C / 145.4 - 147.2 °F

#### **Softening Point**

No data available

#### **Boiling Point/Range**

No information available

#### **Flash Point**

No information available

**Method -** No information available

#### **Evaporation Rate**

Not applicable

Solid

#### **Flammability (solid,gas)**

No information available

#### **Explosion Limits**

No data available

#### **Vapor Pressure**

No data available

#### **Vapor Density**

Not applicable

Solid

#### **Specific Gravity / Density**

No data available

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|                                     |                          |
|-------------------------------------|--------------------------|
| <b>Bulk Density</b>                 | No data available        |
| <b>Water Solubility</b>             | 1088 g/L                 |
| <b>Solubility in other solvents</b> | No information available |

## Partition Coefficient (n-octanol/water)

| Component                                | log Pow |
|------------------------------------------|---------|
| Sodium formaldehydesulfoxylate dihydrate | 3.4     |
| Sodium formaldehyde sulfoxylate          | <0.3    |

|                                  |                          |       |
|----------------------------------|--------------------------|-------|
| <b>Autoignition Temperature</b>  | No data available        |       |
| <b>Decomposition Temperature</b> | 51 °C                    |       |
| <b>Viscosity</b>                 | Not applicable           | Solid |
| <b>Explosive Properties</b>      | No information available |       |
| <b>Oxidizing Properties</b>      | No information available |       |

|                          |                       |
|--------------------------|-----------------------|
| <b>Molecular Formula</b> | C H3 Na O3 S . 2 H2 O |
| <b>Molecular Weight</b>  | 154.11                |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

Yes. Contact with acids liberates very toxic gas.

### Chemical Stability

Hygroscopic.

### Possibility of Hazardous Reactions

|                                 |                                          |
|---------------------------------|------------------------------------------|
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur. |
| <b>Hazardous Reactions</b>      | Contact with acids liberates toxic gas.  |

### Conditions to Avoid

Incompatible products. Excess heat. Avoid dust formation. Exposure to moist air or water.

### Incompatible Materials

Strong acids. Oxidizing agent.

### Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides. Formaldehyde.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

### Product Information

ACR14919

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**(a) acute toxicity;**

Oral

Based on available data, the classification criteria are not met

Dermal

No data available

Inhalation

No data available

| Component                       | LD50 Oral       | LD50 Dermal                | LC50 Inhalation |
|---------------------------------|-----------------|----------------------------|-----------------|
| Sodium formaldehyde sulfoxylate | >2 g/kg ( Rat ) | LD50 >= 2000 mg/kg ( Rat ) | -               |

**(b) skin corrosion/irritation;**

No data available

**(c) serious eye damage/irritation;**

No data available

**(d) respiratory or skin sensitization;**

Respiratory

No data available

Skin

No data available

**(e) germ cell mutagenicity;**

Category 2

Possible risk of irreversible effects

**(f) carcinogenicity;**

No data available

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;**

Reproductive Effects

Category 2

May cause harm to the unborn child.

**(h) STOT-single exposure;**

No data available

**(i) STOT-repeated exposure;**

No data available

Target Organs

No information available.

**(j) aspiration hazard;**

Not applicable

Solid

**Other Adverse Effects**

The toxicological properties have not been fully investigated.

**Symptoms / effects, both acute and delayed** No information available.

**Endocrine Disrupting Properties**

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects**

Do not empty into drains.

**Persistence and degradability**

Persistence

Soluble in water, Persistence is unlikely, based on information available.

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|                                          |                |                                      |
|------------------------------------------|----------------|--------------------------------------|
| <b>Bioaccumulative potential</b>         |                | Bioaccumulation is unlikely          |
| <b>Component</b>                         | <b>log Pow</b> | <b>Bioconcentration factor (BCF)</b> |
| Sodium formaldehydesulfoxylate dihydrate | 3.4            | No data available                    |
| Sodium formaldehyde sulfoxylate          | <0.3           | No data available                    |

**Mobility in soil** The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste treatment methods**  
**Waste from Residues/Unused Products** Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO** Not regulated

**Road and Rail Transport** Not regulated

**IATA** Not regulated

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**International Inventories** X = listed

| Component                                | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|------------------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Sodium formaldehydesulfoxylate dihydrate | -         | -    | -   | -     | -    |      | X     | -    | -        |
| Sodium formaldehyde sulfoxylate          | 205-739-4 | X    | X   | X     | X    | X    | X     | X    | KE-20611 |

**National Regulations**

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**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Revision Date**

21-Mar-2025

**Revision Summary**

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**